LAB EXP 7 - VALUE ITERATION UNIQ ID: E0124092

```
import gym
import numpy as np


env = gym.make("Taxi-v3")
print("Number of states:", env.observation_space.n)
print("Number of actions:", env.action_space.n)


V = np.zeros(env.observation_space.n)

gamma = 0.9
theta = 1e-6

while True:
    delta = 0

    for s in range(env.observation_space.n):
        v = V[s]

        action_values = []

        for a in range(env.action_space.n):
            action_value = 0
            for prob, next_state, reward, done in env.P[s][a]:
                action_value += prob * (reward + gamma *
V[next_state])
            action_values.append(action_value)
        V[s] = max(action_values)
        delta = max(delta, abs(v - V[s]))

    if delta < theta:
        break

policy = np.zeros(env.observation_space.n, dtype=int)
for s in range(env.observation_space.n):
    action_values = np.zeros(env.action_space.n)
    for a in range(env.action_space.n):
        for prob, next_state, reward, done in env.P[s][a]:
            action_values[a] += prob * (reward + gamma *
V[next_state])
    policy[s] = np.argmax(action_values)
print("\nOptimal Value Function:")
print(V)

print("\nOptimal Policy (action per state):")
print(policy)
```

```
/usr/local/lib/python3.13/dist-packages/gym/core.py:317:
DeprecationWarning: WARN: Initializing wrapper in old step API which
returns one bool instead of two. It is recommended to set
`new_step_api=True` to use new step API. This will be the default
behaviour in future.
  deprecation(
/usr/local/lib/python3.13/dist-packages/gym/wrappers/step_api_compatib
ility.py:39: DeprecationWarning: WARN: Initializing environment in old
step API which returns one bool instead of two. It is recommended to
set `new_step_api=True` to use new step API. This will be the default
behaviour in future.
  deprecation(

Number of states: 500
Number of actions: 6

Optimal Value Function:
[ 89.47368235  32.82015648  55.26468138  37.5779509    8.43267265
  32.82015648   8.43267168  15.28487198  32.82015745  18.09430369
  55.26468138  21.21589225  12.75638643  18.09430369  12.75638545
  37.5779509  100.52631412  37.57795224  62.51631324  42.86439071
  79.52631412  28.53814083  48.73821324  32.82015581  10.48074778
  37.57795224  10.48074691  18.09430303  28.53814171  15.28487332
  48.73821324  18.09430303  15.2848742   21.21589359  15.28487332
  42.86439071  89.4736827   42.86439192  55.26468192  48.73821264
  42.8643927   12.75638599  24.68432675  15.28487272  24.68432754
  70.57368192  24.68432675  37.57795164  24.68432754  12.75638599
  42.86439192  15.28487272  18.09430502  24.68432675  18.09430423
  48.73821264  48.73821443  79.52631372  48.73821372  55.26468138
  37.57795343  10.48074739  21.21589407  12.75638545  28.53814202
  79.52631372  28.53814131  42.86439138  21.21589478  10.48074739
  37.57795272  12.75638545  21.21589478  28.53814131  21.21589407
  55.26468138  42.86439299  89.47368235  42.86439235  62.51631324
  32.82015809   8.43267265  18.09430466  10.48074691  32.82015809
  89.47368235  32.82015745  48.73821324  18.0943053    8.43267265
  32.82015745  10.48074691  18.0943053   24.68432718  18.09430466
  48.73821324  37.57795369 100.52631412  37.57795312  55.26468192
  79.52631412  28.53814083  48.73821324  32.82015581  10.48074778
  37.57795224  10.48074691  18.09430303  37.57795312  21.21589359
  62.51631324  24.68432554  15.2848742   21.21589359  15.28487332
  42.86439071  89.4736827   42.86439192  70.57368192  48.73821264
  70.5736827   24.68432675  42.86439192  28.53814023  12.75638678
  42.86439192  12.75638599  21.21589299  32.8201578   18.09430423
  55.26468192  21.21589299  18.09430502  24.68432675  18.09430423
  48.73821264  79.52631443  48.73821372  62.51631372  55.26468138
  48.73821443  15.28487381  28.53814131  18.09430369  21.21589478
  62.51631372  21.21589407  32.82015648  28.53814202  15.28487381
  48.73821372  18.09430369  21.21589478  28.53814131  21.21589407
  55.26468138  55.26468299  70.57368235  55.26468235  62.51631324
  42.86439299  12.75638643  24.68432718  15.28487332  24.68432782
```

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70.57368235  24.68432718  37.57795224  24.68432782  12.75638643
42.86439235  15.28487332  24.68432782  32.82015745  24.68432718
62.51631324  48.73821469  79.52631412  48.73821412  70.57368192
37.57795369  10.48074778  21.21589446  12.75638599  28.53814228
79.52631412  28.53814171  42.86439192  21.21589504  10.48074778
37.57795312  12.75638599  21.21589504  28.53814171  21.21589446
55.26468192  42.86439322  89.4736827   42.8643927   62.51631372
70.5736827   24.68432675  42.86439192  28.53814023  12.75638678
42.86439192  12.75638599  21.21589299  42.8643927   24.68432675
70.57368192  28.53814023  18.09430502  24.68432675  18.09430423
48.73821264  79.52631443  48.73821372  79.52631372  55.26468138
62.51631443  21.21589407  37.57795272  24.68432621  15.28487452
48.73821372  15.28487381  24.68432621  37.57795343  21.21589407
62.51631372  24.68432621  21.21589478  28.53814131  21.21589407
55.26468138  70.57368299  55.26468235  70.57368235  62.51631324
55.26468299  18.09430466  32.82015745  21.21589359  18.0943053
55.26468235  18.09430466  28.53814083  32.82015809  18.09430466
55.26468235  21.21589359  24.68432782  32.82015745  24.68432718
62.51631324  62.51631469  62.51631412  62.51631412  70.57368192
48.73821469  15.2848742   28.53814171  18.09430423  21.21589504
62.51631412  21.21589446  32.82015702  28.53814228  15.2848742
48.73821412  18.09430423  28.53814228  37.57795312  28.53814171
70.57368192  55.26468322  70.5736827   55.2646827   79.52631372
42.86439322  12.75638678  24.68432754  15.28487381  24.68432805
70.5736827   24.68432754  37.57795272  24.68432805  12.75638678
42.8643927   15.28487381  24.68432805  32.8201578   24.68432754
62.51631372  48.7382149   79.52631443  48.73821443  70.57368235
62.51631443  21.21589407  37.57795272  24.68432621  10.4807481
37.57795272  10.48074739  18.09430369  48.73821443  28.53814131
79.52631372  32.82015648  15.28487452  21.21589407  15.28487381
42.86439138  70.57368299  42.86439235  89.47368235  48.73821324
55.26468299  18.09430466  32.82015745  21.21589359  12.75638706
42.86439235  12.75638643  21.21589359  32.82015809  18.09430466
55.26468235  21.21589359  18.0943053   24.68432718  18.09430466
48.73821324  62.51631469  48.73821412  62.51631412  55.26468192
48.73821469  15.2848742   28.53814171  18.09430423  15.28487477
48.73821412  15.2848742   24.68432675  28.53814228  15.2848742
48.73821412  18.09430423  21.21589504  28.53814171  21.21589446
55.26468192  55.26468322  55.2646827   55.2646827   62.51631372
42.86439322  12.75638678  24.68432754  15.28487381  18.09430554
55.2646827   18.09430502  28.53814131  24.68432805  12.75638678
42.8643927   15.28487381  32.82015832  42.8643927   32.8201578
79.52631372  48.7382149   62.51631443  48.73821443  89.47368235
37.5779539   10.4807481   21.21589478  12.75638643  21.21589525
62.51631443  21.21589478  32.82015745  21.21589525  10.4807481
37.57795343  12.75638643  28.53814249  37.57795343  28.53814202
70.57368235  42.86439341  70.57368299  42.86439299  79.52631412
55.26468299  18.09430466  32.82015745  21.21589359   8.43267329
32.82015745   8.43267265  15.28487332  55.26468299  32.82015745
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  89.47368235  37.57795224  12.75638706  18.09430466  12.75638643
  37.57795224  62.51631469  37.57795312 100.52631412  42.86439192
  48.73821469  15.2848742   28.53814171  18.09430423  10.48074836
  37.57795312  10.48074778  18.09430423  28.53814228  15.2848742
  48.73821412  18.09430423  15.28487477  21.21589446  15.2848742
  42.86439192  55.26468322  42.8643927   55.2646827   48.73821372
  42.86439322  12.75638678  24.68432754  15.28487381  12.7563873
  42.8643927   12.75638678  21.21589407  24.68432805  12.75638678
  42.8643927   15.28487381  18.09430554  24.68432754  18.09430502
  48.73821372  48.7382149   48.73821443  48.73821443  55.26468235
  37.5779539   10.4807481   21.21589478  12.75638643  15.28487498
  48.73821443  15.28487452  24.68432718  21.21589525  10.4807481
  37.57795343  12.75638643  37.5779539   48.73821443  37.57795343
  89.47368235  42.86439341  55.26468299  42.86439299 100.52631412
  32.82015851   8.43267329  18.0943053   10.48074778  18.09430572
  55.26468299  18.0943053   28.53814171  18.09430572   8.43267329
  32.82015809  10.48074778  32.82015851  42.86439299  32.82015809
  79.52631412  37.57795407  62.51631469  37.57795369  89.4736827 ]

Optimal Policy (action per state):
[4 4 4 4 0 0 0 0 0 0 0 0 0 0 0 0 5 0 0 0 3 3 3 3 0 0 0 0 0 0 0 0 0 0 0
0 3
 0 0 0 0 0 0 0 2 2 2 2 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 2 2 2 2 0 0 0 0
0 0
 0 0 0 2 0 0 0 0 0 0 4 4 4 4 0 0 0 0 0 0 0 0 0 5 0 0 1 1 1 1 0 0 0 0 0
0 0
 0 0 0 0 0 1 0 0 0 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 2 2
2 2
 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 2 2 2 2 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0
0 1
 1 1 1 0 0 0 0 0 0 0 0 0 1 0 0 1 1 1 1 2 2 2 2 0 0 0 0 2 2 2 2 1 2 0 2
1 1
 1 1 2 2 2 2 3 3 3 3 2 2 2 2 1 2 3 2 3 3 3 3 2 2 2 2 3 3 3 3 2 2 2 2 3
2 3
 2 3 3 3 3 2 2 2 2 3 3 3 3 0 0 0 0 3 2 3 0 3 3 3 3 1 1 1 1 3 3 3 3 0 0
0 0
 3 1 3 0 1 1 1 1 1 1 1 1 0 0 0 0 1 1 1 1 1 1 0 1 1 1 1 1 2 2 2 2 1 1 1
1 2
 2 2 2 1 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2
1 1
 1 1 0 0 0 0 1 2 1 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 1 1 1 0 1 1 1 1 1
1 1
 1 4 4 4 4 1 1 1 1 1 1 5 1 1 1 1 1 2 2 2 2 1 1 1 1 2 2 2 2 1 2 1 2 1 1
1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 1 1 1 1 4 4 4 4 1 2 1
5 1
 1 1 1 1 1 1 1 1 1 1 1 3 3 3 3 1 1 1 3]
```